Xin Liu

Email: liuxin.optics@gmail.com
Website: liux2018.github.io
ORCID: 0000-0002-4371-018X

Postdoctoral Researcher in Computational Optics

Department of Electrical and Computer Engineering
The University of Hong Kong
Hong Kong SAR, China

Research profile Computational optics researcher developing physically grounded models and algorithms to simulate and shape light for imaging and sensing.

Research areas Computational Optics; Optical Simulation; Light Shaping; Computational Imaging.

Academic Appointments

04/2024 – Present **Postdoctoral Researcher** Hong Kong SAR, China
Department of Electrical and Computer Engineering, The University of Hong Kong (HKU)

01/2024 – 03/2024 **Research Assistant** Hangzhou, China
State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)

Education

09/2018 – 12/2023 **Ph.D. in Optical Engineering** Hangzhou, China
State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)
Dissertation: *Research on the principles and key issues of point spread function engineering*

09/2014 – 06/2018 **B.E. in Optoelectronic Information Science and Engineering** Guangzhou, China
School of Physics and Optoelectronics, South China University of Technology (SCUT)

Journal Publications (Selected)

*equal contribution; ✉corresponding author.

[11] **Retained accuracy with reduced precision in wave propagation modeling**. Xin Liu and Yifan Peng. *Photonics Research*, 2026.

[10] **Learned off-aperture encoding for wide field-of-view RGBD imaging**. Haoyu Wei, Xin Liu, Yuhui Liu, Qiang Fu, Wolfgang Heidrich, Edmund Y. Lam, and Yifan Peng. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025.

[9] **In situ fully vectorial tomography and pupil function retrieval of tightly focused fields**. Xin Liu, Shijie Tu, Yiwen Hu, Yifan Peng, Yubing Han, Cuifang Kuang, Xu Liu, and Xiang Hao. *Nature Communications*, 2025.

[8] **Diffraction modeling between arbitrary planes using angular spectrum rearrangement**. Yiwen Hu*, Xin Liu*✉, Xu Liu, and Xiang Hao✉. *Optica*, 2025. **[Monthly Top Downloads in January]**

[7] **Modeling off-axis diffraction with the least-sampling angular spectrum method**. Haoyu Wei*, Xin Liu*, Xiang Hao, Edmund Y. Lam, and Yifan Peng. *Optica*, 2023. **[Monthly Top Downloads in July & August]**

[6] **Fast generation of arbitrary optical focus array**. Xin Liu, Yiwen Hu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao. *Optics and Lasers in Engineering*, 2023.

[5] **Investigating deep optics model representation in affecting resolved all-in-focus image quality and depth estimation fidelity**. Xin Liu*, Linpei Li*, Xu Liu, Xiang Hao, and Yifan Peng. *Optics Express*, 2022.

[4] **Calibration of phase-only liquid-crystal spatial light modulators by diffractogram analysis**. Xin Liu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao. *Optics and Lasers in Engineering*, 2022.

[3] **Accurate background reduction in adaptive optical three-dimensional stimulated emission depletion nanoscopy by dynamic phase switching**. Shijie Tu, Xin Liu, Difu Yuan, Wenli Tao, Yubing Han, Yan Shi, Yanghui Li, Cuifang Kuang, Xu Liu, Yufeng Yao, Yesheng Xu, and Xiang Hao. *ACS Photonics*, 2022. **[On the Cover]**

[2] **Generation of arbitrary longitudinal polarization vortices by pupil function manipulation**. Xin Liu, Yifan Peng, Shijie Tu, Jun Guan, Cuifang Kuang, Xu Liu, and Xiang Hao. *Advanced Photonics Research*, 2021. **[On the Cover]**

[1] **Aberrations in structured illumination microscopy: A theoretical analysis**. Xin Liu, Shijie Tu, Yan Xu, Hongya Song, Wenjie Liu, Qiulan Liu, Cuifang Kuang, Xu Liu, and Xiang Hao. *Frontiers in Physics*, 2020.

Awards and Honors

2025 **Outstanding Young Rising Star Award (Ranked 1st of 20)**
The 5th International Conference on Computational Imaging (CITA 2025)
Chinese Society for Optical Engineering

2024 **Honorable Mention**
The 4th International Conference on Computational Imaging (CITA 2024)
Chinese Society for Optical Engineering

2022 **Scholarship for Innovation and Entrepreneurship**
Zhejiang University

2021, 2022 **Merit Graduate Student**
Zhejiang University

2020–2022 **Junshi Chen Scholarship**
Zhejiang University
Awarded in 2020, 2021, and 2022.

2020, 2021 **Outstanding Graduate Student Cadre**
Zhejiang University

2019–2022 **Outstanding Graduate Student**
Zhejiang University
Awarded in 2019, 2021, and 2022.

Talks

09/2025 **Exploring advanced imaging with computational optics** — Suzhou, China
The 5th International Conference on Computational Imaging (CITA 2025)

09/2024 **Free-space wave propagation modeling with the least-sampling angular spectrum method** — Xiamen, China
The 4th International Conference on Computational Imaging (CITA 2024)

05/2024 **Exploring wave propagation modeling in free space** — Thuwal, Saudi Arabia
Guest lecture, King Abdullah University of Science and Technology

Research Mentoring

2022 – 2025 **Learned wide-angle imaging with optical modeling and optimization** — Ph.D. thesis
Haoyu Wei
The University of Hong Kong

2021 – 2025 **Characterization of optical fields with computational methods** — Ph.D. thesis
Yiwen Hu
Zhejiang University

2023 **Deep stereo optics imaging for high-fidelity depth estimation** — Master's thesis
Yuhui Liu
The University of Hong Kong

2022 **Full vectorial manipulation of tightly focused optical fields** — Bachelor's thesis
Haiwei Wang
Zhejiang University

2022 **Numerical simulation of vectorial fields based on vectorial diffraction theory** Bachelor's thesis
Yunpeng Wang
Changchun University of Science and Technology

Academic Service

Journal Reviewer Optica; Advanced Functional Materials; Laser & Photonics Reviews; Advanced Science; Optics Letters; Optics Express; JOSA A; JOSA B; Applied Optics; Journal of Microscopy; Optics Communications

Conference Reviewer SIGGRAPH; SIGGRAPH Asia; International Conference on Computational Photography (ICCP)

Guest Editor Computational Methods for Advanced Optical Imaging, Photonics, MDPI, 2026

Program Chair HKU-KAUST Joint Postgraduate Workshop on Computational Imaging, 2025

Program Committee International Conference on Computational Photography (ICCP), 2026
The 5th International Conference on Computational Imaging (CITA), 2025

Patents

2022 **A method and device for calibration of phase-only spatial light modulators**
Xiang Hao, Xin Liu, Yubing Han, Cuifang Kuang, and Xu Liu
Patent No. CN112697401A

2020 **A method for point spread function reconstruction**
Xiang Hao, Xin Liu, Cuifang Kuang, and Xu Liu
Patent No. CN110133849B

2020 **A two-dimensional rapid beam scanning module for confocal fluorescence microscopy**
Xiang Hao, Xin Liu, Shijie Tu, and Xu Liu
Patent No. CN109491065A

2020 **A device for common-path beam modulation in imaging and lithography systems**
Xu Liu, Xin Liu, Xiang Hao, Cuifang Kuang, and Haifeng Li
Patent No. CN110568650B

2020 **A device for generating super-resolution optical focus arrays**
Xiang Hao, Xin Liu, Cuifang Kuang, and Xu Liu
Patent No. CN110568731B

2020 **A multicolor super-resolution microscope with automatic alignment capability**
Xiang Hao, Shijie Tu, Xin Liu, and Xu Liu
Patent No. CN109358030B